DEFINE PROBLEM STATEMENT

Project Name: EduGenie

Technology: Google Gemini Generative AI

Project Type: AI-Powered Learning Assistant

1. INTRODUCTION

Students have to study a large amount of educational content for assignments, examinations and revision. Reading lengthy textbooks, notes and other study materials can take a lot of time. Students may also find it difficult to identify the most important information from large amounts of content.

EduGenie is designed as an AI-powered learning assistant that helps students understand and revise educational content more efficiently.

2. PROBLEM IDENTIFICATION

The main problem is that students spend a considerable amount of time reading and manually summarizing lengthy educational materials.

The major difficulties faced by students include:

- Large amounts of study material.
- Difficulty identifying important information.
- Time-consuming manual note preparation.
- Difficulty understanding complex topics.
- Limited time for examination preparation.

• Repeated reading during revision.

3. PROBLEM STATEMENT

Students require a simple and efficient way to understand and revise lengthy educational content. Manually reading, understanding and summarizing large amounts of study material consumes significant time and effort.

Therefore, there is a need for an AI-powered learning assistant that can process educational content and generate concise, clear and meaningful summaries to support students in learning and revision.

4. TARGET USERS

The primary users of EduGenie are:

• School students.

• College students.

• Students preparing for examinations.

• Self-learners.

• Students who need quick revision material.

5. PROPOSED SOLUTION

EduGenie uses Google Gemini Generative AI to assist students with educational content.

The user provides educational text to the system. EduGenie processes the content using the Gemini model and generates a concise and understandable response.

This reduces the effort required to manually summarize lengthy educational material.

6. PROJECT OBJECTIVE

The main objective of EduGenie is to develop a Google Gemini powered learning assistant that helps students convert lengthy educational content into concise, clear and useful study material.

7. EXPECTED OUTCOME

The expected outcome of the project is an easy-to-use AI learning assistant that:

- Accepts educational content from the user.
- Processes the content using Google Gemini.
- Generates concise and meaningful summaries.
- Helps students understand important information.
- Supports faster revision and learning.

8. FUTURE SCOPE

The EduGenie project can be extended in the future with:

- AI-based question answering.

- Personalized learning assistance.
- Quiz generation.
- Flashcard generation.
- Revision planning.
- Topic-wise learning support.
- Student performance tracking.

9. CONCLUSION

EduGenie aims to reduce the difficulty of studying lengthy educational content by providing AI-assisted learning support. By using Google Gemini Generative AI, the project provides a foundation for faster, simpler and more effective learning and revision.